

WorkInSync Access Card Management Integration - API based

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Ver No.	Author	Approved by	Approved Date	Description
1.0	Rahul Agrawal	Ujjwal Trivedi	28-Jun-2022	Initial Doc
1.1	Binoy Dedhia	Ujjwal Trivedi	29-Nov-2022	Updated version (with actual API signature)
1.2	Aditya Dutta	Ujjwal Trivedi	19-Jul-2024	Addition of RFID

Purpose

This document explains the process and data required for integrating WorkInSync system with client's own/third party access management vendors using APIs.

Prerequisite

A Client specific username and password will be shared with the client.

Description

Data Integration is to facilitate regular syncing of data between Client/vendor systems and WorkInSync via rest-based APIs. The following services from WorkInSync would be involved:

API Gateway: The API gateway will facilitate all interactions between the client service and the WorkInSync services. Current assumption is that all APIs will be client/vendor initiated interactions. Key aspects of this gateway like throttling, fault tolerance, secure data exchange mechanism, error logging and communication need to be discussed in further detail.

Authentication

Workinsync API interface uses a Bearer token mechanism for APIs provided to the clients. A Client specific username and password will be shared with the client. Before calling any other APIs, GetAuthToken API needs to be called to receive an access token. On success, a short lived access token will be received in response to this API call. This access token in turn needs to be passed in the header of each API call as a 'bearer' token. If at any point the current authorization token expires, the 'Get AuthToken' API needs to be called again to receive a fresh access token.

Authentication:

URL: {baseUrl}/auth/token

Header: Authorization: Basic <base64(username:password)>

--form 'grant_type ="client_credentials"

Response:

Success:

```
{
  "access_token": "eyJhbGciOiJSUzI1Ni.....DyCIIJYEMA4g",
  "token_type": "bearer",
  "expires_in": 172799,
```

```
"scope": "write",
"jti": "5feee94d-f3a7-4939-bea6-a630b0f87e7e"
}
```

API

1. **AccessCardCheckIn:** CheckIn to an existing booking

Request format from Access control system				
Field Name	Format	Field Mapping	Length	Constraints
filter*	String	EmployeeID or EmployeeName or EmployeeEmailID of the Employee scanning so that we can check the booking for them	Max: 50	Should be unique within each employee
officeName*	String	Office name for where the action was performed	Max: 100	
bookingStatus*	String	Defines the action to be taken i.e. checkin in this case	Max: 50	Send "SIGNED_IN" so that we know we have to check in or "SIGNED_OUT" so that we know we have to checkout
epochTime*	Long	Time Stamp at which the above activity has been performed		Date format will be in Epoch format in milliseconds
premiseId*	String	Type of booking that is requested	Max: 50	Returns for all booking types if nothing is mentioned. Possible Values OFFICE,

				PARKING, PARKING_TWO , PARKING_FOU R, MEALS, MEETING
readerId	String	The unique ID associated with the device at the entry point.	Max: 50	Reader ID to office ID mapping would be required as an one time activity
rfid	string	Unique identification for a card No.	Max: 50	Would need to create a mapping for RFID with corresponding employees. Filter can be skipped if this is passed

Response				
Field Name	Format	Field Mapping	Length	Description
status	Integer	API status code	Max : 4	Represents API success/failure
data	UUID	Booking Id	Max : 50	
message	String	Success/Failure message of API		

Sample API request and response

Prerequisites

- We're assuming that the scanner device will communicate with WorkInSync system in real time
- We will need clients to provide mentioned information about the vendors that will consume the REST APIs via API gateway. It is required for registering clients for authentication and authorization on the gateway
- Without proper Auth, external 3rd parties can't access our internal APIs

Note:

- If the createBookingIfNotPresent flag is true, we will create an office booking for employees who do not have an office booking based on scan time and end time.
- This is a sample API and it may change

baseUrl:- <https://api.moveinsync.in/integration/>

URL: POST {baseUrl}/integration/bookings/ci-co

Header: Authorization: Bearer <Auth token> → generated from above authentication API

Payload:

```
{
  "bookingStatus": "SIGNED_IN",
  "epochTime": 1668428940000,
  "filter": "qa134@mis.com",
  "premiseId": "",
  "readerId": "wertyhbf1239df",
  "officeName": "",
  "rfid": ""
}
```

Instructions regarding the payload:

1. BookingStatus can be SIGNED_IN/SIGNED_OUT. SIGNED_IN status will check-in into a booking or create a booking and check-in into it based on the property. SIGNED_OUT will sign out the employee from the existing booking.
2. epochTime is the time of check-in/check-out.
3. filter accepts either emailId/phone number/emplID. If there is rfid associated with the employee, then it can be passed in the rfid variable by leaving the filter variable as blank.
4. ReaderId or premiseId or officeName need to be passed for identifying the office.

Sample Response:

```
{
  "status": 200,
  "data": "832dbe42-f3eb-48cd-91d8-8c81ccf24ca7",
  "message": "Check-in successful"
}
```

Instructions regarding the response:

1. API status code will be 200 every time. The status in the response will be changed to 1001 if there is any internal failure. And a message will be updated according to the failure status.
2. Status code will be 200 if the check-in/check-out is successful.

Different types of API responses:

i:

API response status code will be 200

```
{  
  "status": 200,  
  "data": "832dbe42-f3eb-48cd-91d8-8c81ccf24ca7",  
  "message": "Check-in successful"  
}
```

ii:

API response status code will be 200

```
{  
  "status": 200,  
  "data": "832dbe42-f3eb-48cd-91d8-8c81ccf24ca7",  
  "message": "Check-out Successful"  
}
```

iii:

API response status code will be 401 if the Authorization code is not passed.

```
{  
  "error": "unauthorized",  
  "error_description": "Full authentication is required to access this resource"  
}
```

iv:

If there is any internal failure:

API response status code will be 200

```
{  
  "status": 1001,  
  "data": null,  
  "message": "Some problem processing your request"  
}
```

Sample cURL

Generated with the credentials for access token -

```
curl --location 'https://api.moveinsync.in/auth/token' \  
--header 'Authorization: Basic  
MDU4NmM4ZDMtNDE2Ni00OGMyLTg1OTItZWJhNDQwZWY3ZWU3OnloT1pRb0tmbVpRRTI  
2eg==' \  
--form 'grant_type="client_credentials"'
```

Sample cURL for “bookings/ci-co” Api

```
curl --location 'https://api.moveinsync.in/integration/bookings/ci-co' \
--header 'accept: */*' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.eyJhdWQiOiI0ZXh0ZXJuYWwtaW50ZWdyYXRpb
24iXSwic2NvcGUiOiJpdGUiXSwiZXhwljoxNzlyNjY5Nzk3LCJqdGkiOiJxdnNUcDJkdHY1S
WV1V0p0VIZTVW9od0N0TmciLCJjbGllbnRfaWQiOiJlNTg2YzhkMy00MTY2LTQ4YzltODU5
Mi1lYmE0NDBlYzRhZTcifQ.OBflwaTZqSQ7ZJ110GnHWBteclOI7oEFnuGcb-GoZipgcwdul6Y
KI6Gqz-yicsvpp6HswTb1PznFHuwolRDDzY1ocqQCOPR2MzSRsu3ybXwxWtauEdylRxBDT
PcVDxDAS6d8wV_5zBjc6mfA2-NgqD2aXr3WzLcA-6R-MLTH26NbeOMx5wi1V8EaKF1rbypN
FUKK9S2oT-rNUT-BkfRYKY0g7VBMi7Ti9F6-VeGy3BM1jFCoG_EOgMaEQwG5fWGjzHgRI3
ARpHCt2v2BWS_uTAECjKUnaNw_LQDAuqztRXA_9KroHtqFj_YAiHbswgigESmKsU2DM8qA
-Uh4Q_muhA' \
--data '{
  "rflid": "11228899",
  "bookingStatus": "SIGNED_IN",
  "epochTime": 1722493865345,
  "officeName": "Prestige Trade Center",
  "premiseId": "864dd2c3-a1e8-4e2c-a646-cae76b36d5a2"
}'
```

Payload

Payload:

```
{
  "bookingStatus": "SIGNED_IN",
  "epochTime": 1668428940000,
  "filter": "",
  "premiseId":
"7107f124-cc4d-4495-8121-81e7f6594252/864dd2c3-a1e8-4e2c-a646-cae76b36d5a2",
  "readerId": "",
  "officeName": "Prestige Trade Center",
  "rflid": "2959067"
}
```

Response

Success

```
{  
  "status": 200,  
  "data": "832dbe42-f3eb-48cd-91d8-8c81ccf24ca7",  
  "message": "Check-in successful"  
}
```

Failure

If booking is not found:

```
{  
  "status": 1001,  
  "data": null,  
  "message": "Booking is not present for rfid: 123"  
}
```

Other error code and and messages please refer above